

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	261498	50.039036
1	[Git, Change, Create]	92826	17.762750
2	[Git, Change, Delete]	58957	11.281736
3	[Git, Change, Rename]	24301	4.650126
4	[File, Git]	16199	3.099765
5	[Git, Commit]	15955	3.053074
6	[Java, ByteCode, Parameter]	9236	1.767358
7	[Java, ByteCode, Member, Method]	8935	1.709760
8	[Java, ByteCode, Bound]	6321	1.209557
9	[Java, Value, ByteCode, Annotation]	5389	1.031214
10	[Java, ByteCode, Bound, ParameterizedType]	4895	0.936684
11	[Xml, Element]	2416	0.462314
12	[Java, ByteCode, Member, Field]	2202	0.421364
13	[Xml, Text]	1622	0.310378
14	[Java, ByteCode, Member, Constructor, Method]	1583	0.302915
15	[Java, ByteCode, Bound, WildcardType]	970	0.185615
16	[Java, ByteCode, Member, Method, Lambda]	851	0.162843
17	[Java, ByteCode, Bound, TypeVariable]	848	0.162269
18	[Type, File, Java, ByteCode, JavaType]	690	0.132035
19	[Type, File, Java, ByteCode, Class]	660	0.126295
20	[Json, Key]	544	0.104097
21	[Value, Json, Scalar]	534	0.102184
22	[Git, Change, Copy]	481	0.092042
23	[Type, File, Java, ByteCode, ResolvedDuplicate...	477	0.091276
24	[Java, ByteCode, GenericDeclaration, Member, M...	466	0.089172
25	[Java, Value, ByteCode, Primitive]	329	0.062956
26	[Type, File, Java, ByteCode, ExternalType]	313	0.059894
27	[Author, Git, Person]	304	0.058172
28	[Committer, Git, Person]	247	0.047265
29	[Maven, Dependency]	195	0.037314

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

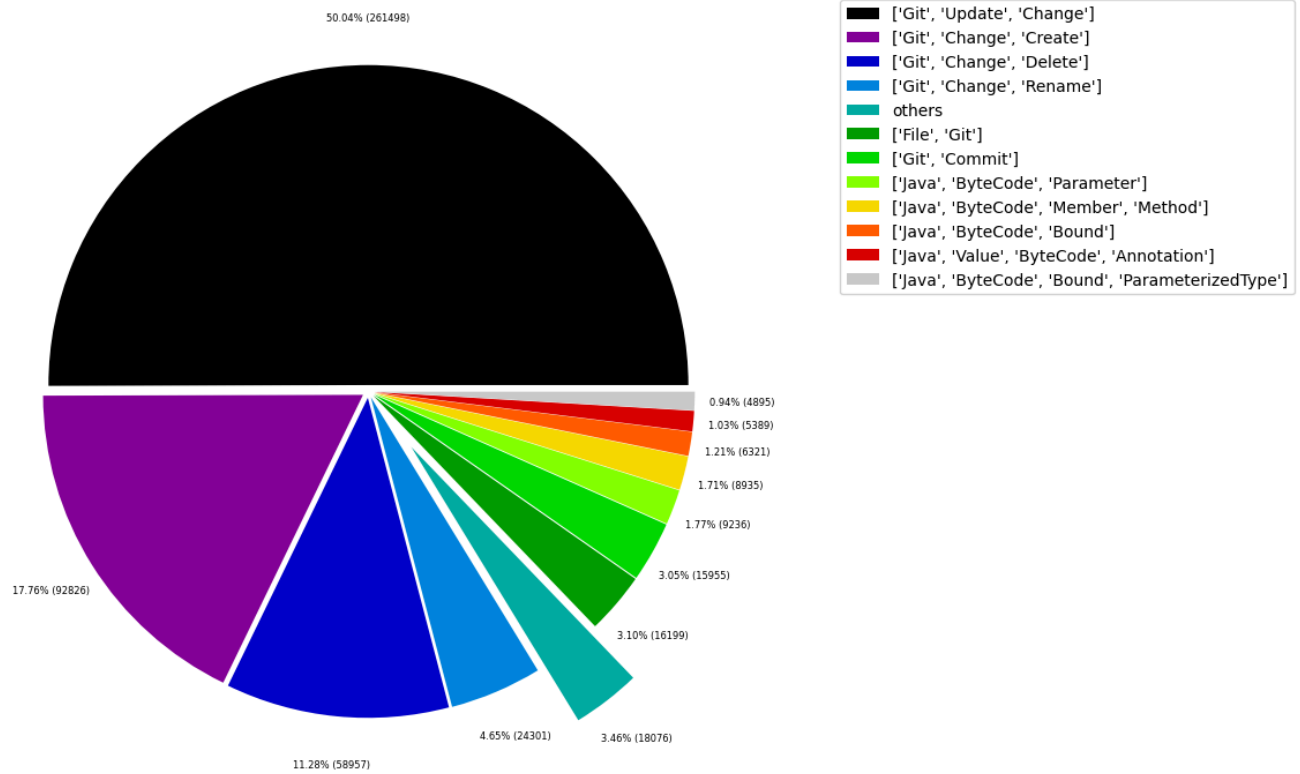


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000191
1	[Git, Branch]	1	0.000191
2	[Repository, File, Git]	1	0.000191
3	[File, Json]	2	0.000383
4	[File]	3	0.000574
5	[Type, File, Java, ByteCode, GenericDeclaratio...	4	0.000765
6	[Maven, Exclusion]	4	0.000765
7	[Value, Array, Json]	8	0.001531
8	[Type, File, Java, ByteCode, Throwable, Extern...	8	0.001531
9	[Java, ByteCode, GenericDeclaration, Member, C...	9	0.001722
10	[Type, File, Java, ByteCode, Throwable, Resolv...	11	0.002105
11	[Maven, Scm]	11	0.002105
12	[File, Maven, Xml, Pom, Document]	11	0.002105
13	[Type, File, Java, ByteCode, Void]	11	0.002105
14	[Java, ManifestSection]	11	0.002105
15	[File, Java, Manifest]	11	0.002105
16	[File, Artifact, Jar, Archive, Zip, Java]	11	0.002105
17	[File, Java, ServiceLoader]	12	0.002296
18	[Maven, PluginExecution]	13	0.002488
19	[File, Java, Properties]	13	0.002488
20	[Maven, ExecutionGoal]	13	0.002488
21	[Type, File, Java, ByteCode, Enum]	17	0.003253
22	[jQAssistant, Rule, Concept]	19	0.003636
23	[Maven, Configuration]	20	0.003827
24	[Maven, Plugin]	20	0.003827
25	[Xml, Attribute]	22	0.004210
26	[Type, File, Java, ByteCode, PrimitiveType]	39	0.007463
27	[Type, File, Java, ByteCode, Annotation]	42	0.008037
28	[Type, File, Java, ByteCode, Class, Throwable]	42	0.008037
29	[Xml, Namespace]	44	0.008420

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

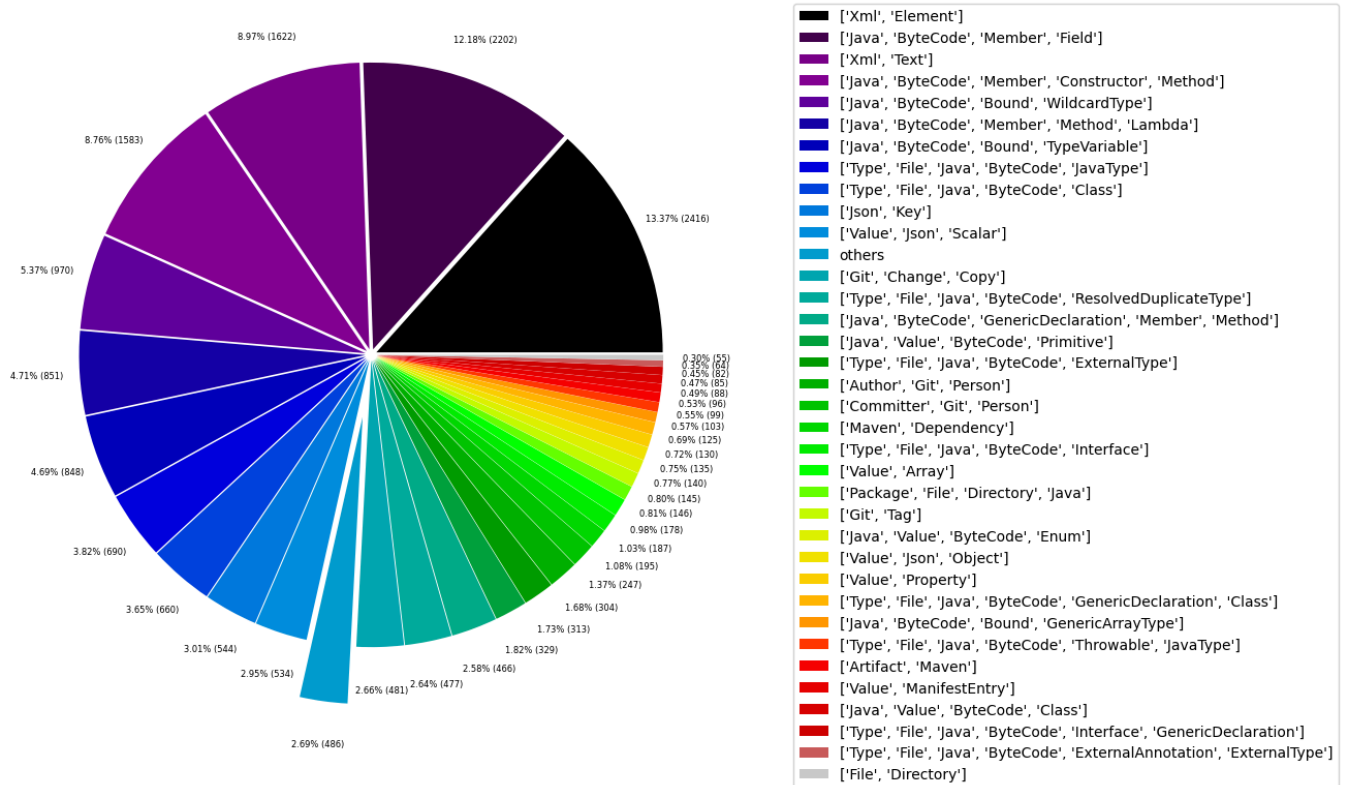


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

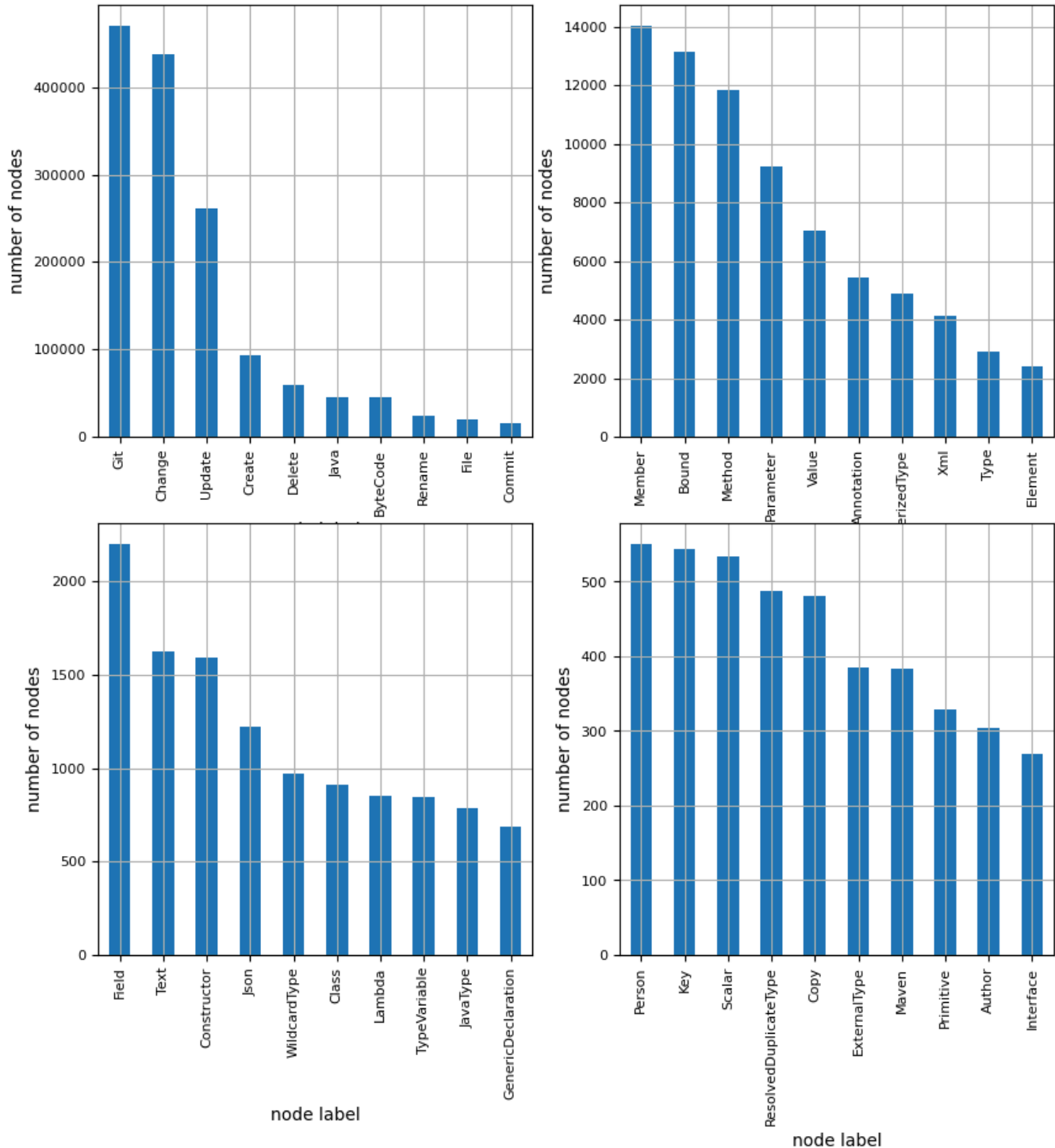
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	470915	90.112096
1	Change	438063	83.825691
2	Update	261498	50.039036
3	Create	92826	17.762750
4	Delete	58957	11.281736
5	Java	45489	8.704563
6	ByteCode	45285	8.665526
7	Rename	24301	4.650126
8	File	19387	3.709806
9	Commit	15955	3.053074
10	Member	14046	2.687777
11	Bound	13137	2.513835
12	Method	11844	2.266413
13	Parameter	9236	1.767358
14	Value	7016	1.342549
15	Annotation	5431	1.039251
16	ParameterizedType	4895	0.936684
17	Xml	4115	0.787427
18	Type	2923	0.559332
19	Element	2416	0.462314
20	Field	2202	0.421364
21	Text	1622	0.310378
22	Constructor	1592	0.304638
23	Json	1223	0.234028
24	WildcardType	970	0.185615
25	Class	912	0.174516
26	Lambda	851	0.162843
27	TypeVariable	848	0.162269
28	JavaType	789	0.150979
29	GenericDeclaration	686	0.131270
30	Person	551	0.105437
31	Key	544	0.104097
32	Scalar	534	0.102184
33	ResolvedDuplicateType	488	0.093381
34	Copy	481	0.092042
35	ExternalType	385	0.073672
36	Maven	383	0.073289
37	Primitive	329	0.062956
38	Author	304	0.058172
39	Interface	269	0.051475

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1621729

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	438063	27.012096
1	MODIFIES	438063	27.012096
2	UPDATES	261498	16.124642
3	CREATES	117608	7.252013
4	DELETES	83258	5.133903
5	COMMITTED	31910	1.967653
6	RENAMES	24301	1.498462
7	INVOKES	22459	1.384880
8	HAS_PARENT	19403	1.196439
9	OF_TYPE	18564	1.144704
10	DEPENDS_ON	17816	1.098581
11	HAS_FILE	16199	0.998872
12	HAS_COMMIT	15955	0.983827
13	DECLARES	14394	0.887571
14	OF_RAW_TYPE	13189	0.813268
15	HAS	9866	0.608363
16	HAS_NEW_NAME	9446	0.582465
17	RETURNS	8167	0.503598
18	READS	6080	0.374909
19	HAS_ACTUAL_TYPE_ARGUMENT	5811	0.358321
20	ANNOTATED_BY	5388	0.332238
21	OF_GENERIC_TYPE	4376	0.269835
22	CHANGED_TOGETHER_WITH	4036	0.248870
23	SIMILAR	3419	0.210824
24	CONTAINS	3227	0.198985
25	RESOLVES_TO	2621	0.161618
26	RETURNS_GENERIC	2601	0.160384
27	HAS_FIRST_CHILD	2416	0.148977
28	HAS_LAST_CHILD	2416	0.148977
29	HAS_ELEMENT	2394	0.147620

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

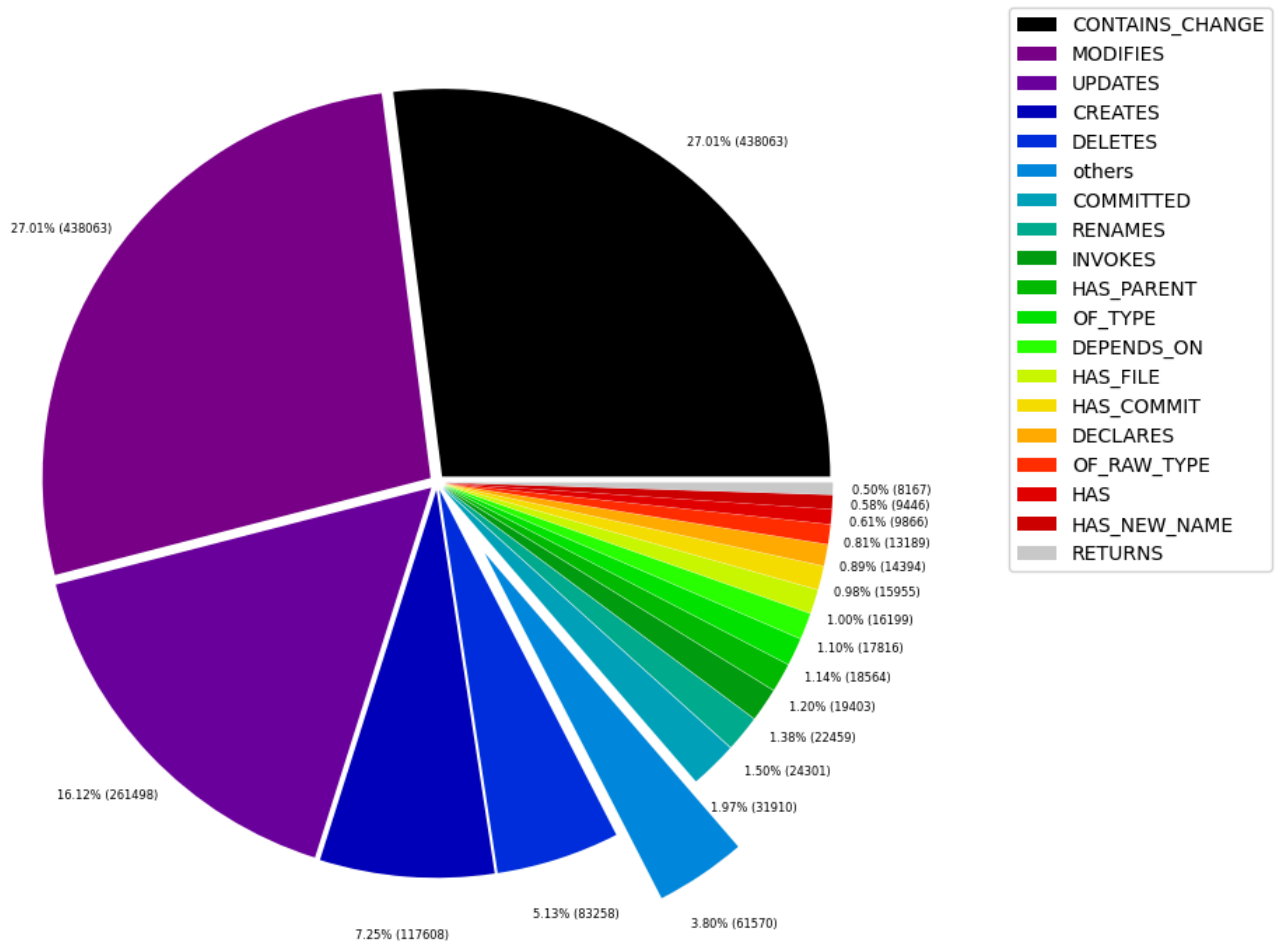


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

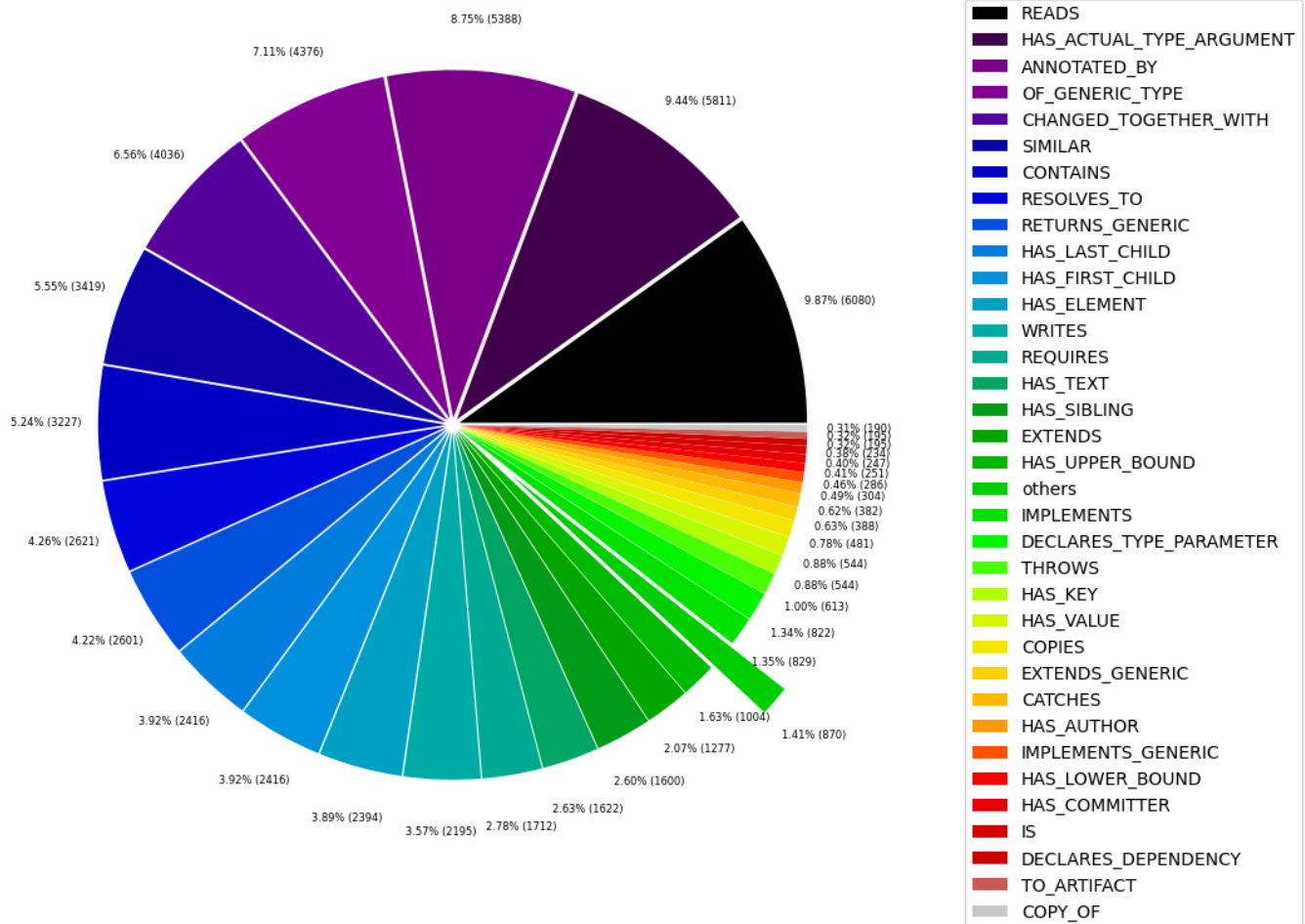
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000062
1	HAS_BRANCH	1	0.000062
2	HAS_HEAD	2	0.000123
3	EXCLUDES	4	0.000247
4	THROWS_GENERIC	5	0.000308
5	HAS_SCM	11	0.000678
6	HAS_ROOT_ELEMENT	11	0.000678
7	DESCRIBES	11	0.000678
8	HAS_GOAL	13	0.000802
9	HAS_EXECUTION	13	0.000802
10	INCLUDES_CONCEPT	19	0.001172
11	IS_ARTIFACT	20	0.001233
12	HAS_CONFIGURATION	20	0.001233
13	USES_PLUGIN	20	0.001233
14	OF_NAMESPACE	22	0.001357
15	HAS_ATTRIBUTE	22	0.001357
16	REQUIRES_TYPE_PARAMETER	26	0.001603
17	REQUIRES_CONCEPT	28	0.001727
18	DECLARES_NAMESPACE	44	0.002713
19	HAS_DEFAULT	53	0.003268
20	HAS_COMPONENT_TYPE	103	0.006351
21	CONTAINS_VALUE	131	0.008078
22	HAS_TAG	145	0.008941
23	ON_COMMIT	145	0.008941
24	COPY_OF	190	0.011716
25	TO_ARTIFACT	195	0.012024
26	DECLARES_DEPENDENCY	195	0.012024
27	IS	234	0.014429
28	HAS_COMMITTER	247	0.015231
29	HAS_LOWER_BOUND	251	0.015477

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberO
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	261498		15955
1	[Git, Update, Change]	UPDATES	[File, Git]	261498		261498
2	[Git, Update, Change]	MODIFIES	[File, Git]	261498		261498
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	92826		15955
4	[Git, Change, Create]	MODIFIES	[File, Git]	92826		92826
5	[Git, Change, Create]	CREATES	[File, Git]	92826		92826
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	58957		15955
7	[Git, Change, Delete]	DELETES	[File, Git]	58957		58957
8	[Git, Change, Delete]	MODIFIES	[File, Git]	58957		58957
9	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	24301		15955
10	[Git, Change, Rename]	DELETES	[File, Git]	24301		24301
11	[Git, Change, Rename]	CREATES	[File, Git]	24301		24301
12	[Git, Change, Rename]	RENAMES	[File, Git]	24301		24301
13	[Git, Change, Rename]	MODIFIES	[File, Git]	24301		24301
14	[Git, Commit]	HAS_PARENT	[Git, Commit]	19392		15955
15	[Repository, File, Git]	HAS_FILE	[File, Git]	16199		1
16	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	15955		1
17	[Author, Git, Person]	COMMITTED	[Git, Commit]	15955		304
18	[Committer, Git, Person]	COMMITTED	[Git, Commit]	15955		247
19	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	13380		8905
20	[File, Git]	HAS_NEW_NAME	[File, Git]	9446		16199
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	5458		8905
22	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	5321		8905
23	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalAnnotatio...	5035		5389
24	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	4123		9236
25	[Java, ByteCode, Parameter]	ANNOTATED_BY	[Java, Value, ByteCode, Annotation]	3821		9236
26	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	3341		16199
27	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	2826		4895
28	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2394		2416
29	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	2297		6321

Graph Density

```
total_number_of_nodes (vertices): 522588
total_number_of_relationships (edges): 1621729
-> total directed graph density: 5.938274210163616e-06
-> total directed graph density in percent: 0.0005938274210163616
```